# Tutorial 11

# Trees and networks

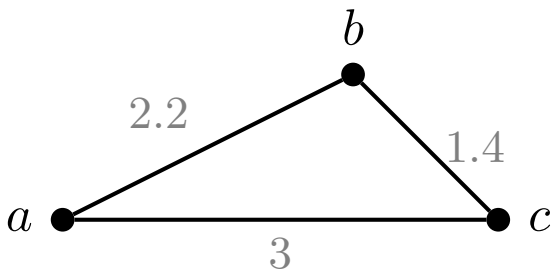
**Homework**

1. Provide detailed solutions to *three* selected parts of Question 3.
2. Provide detailed solutions to Question 5(i)–(iv) for *two* selected graphs from Question 5.

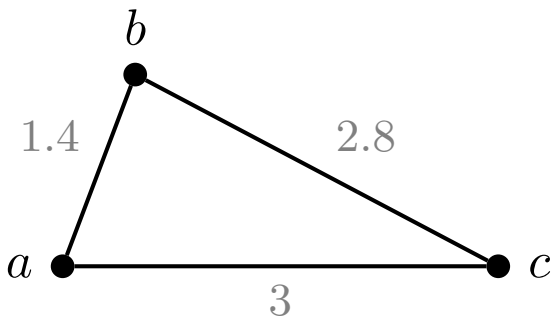
**Question 1.** For each of the following triangles

- Perform Torricelli's Construction to find the Fermat point. It is recommended that you use a protractor (or a compass) and a ruler.
- Compute the total length of the shortest network connecting the vertices of the triangle.

1.




2.



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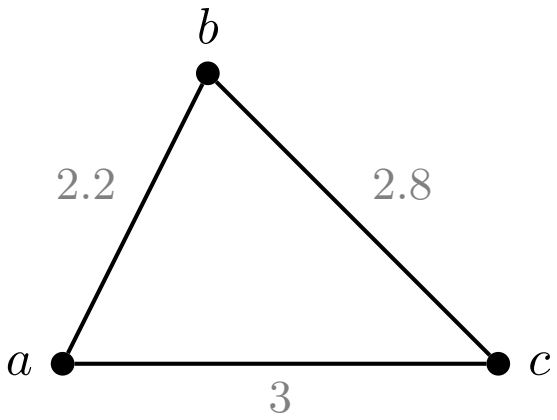
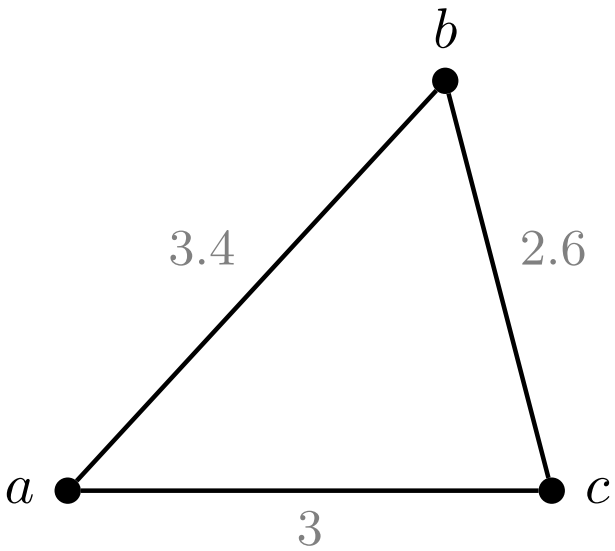



4.



*Solution.*

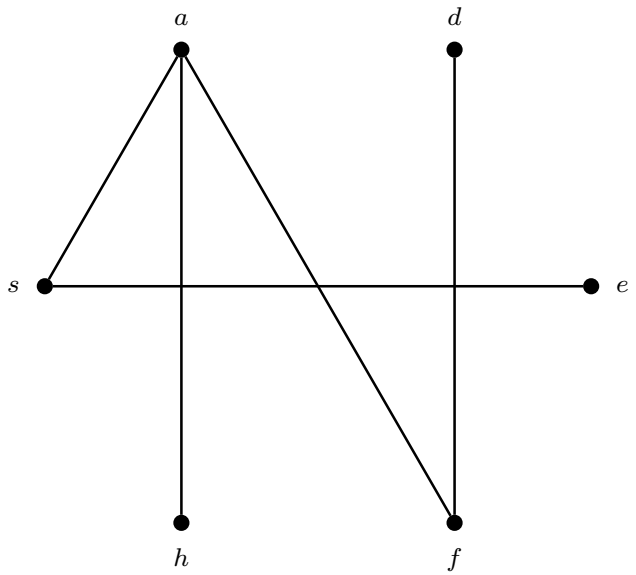
1. 3.633
2. 4.044
3. 4.625
4. 5.206

**Question 2.** Alice must visit clients in six cities next month and needs to minimize her driving mileage. The table below lists the distances between these cities. Use the mTSP Algorithm to find a good travel plan if she must start and end her trip in Dallas. Compute the total distance.

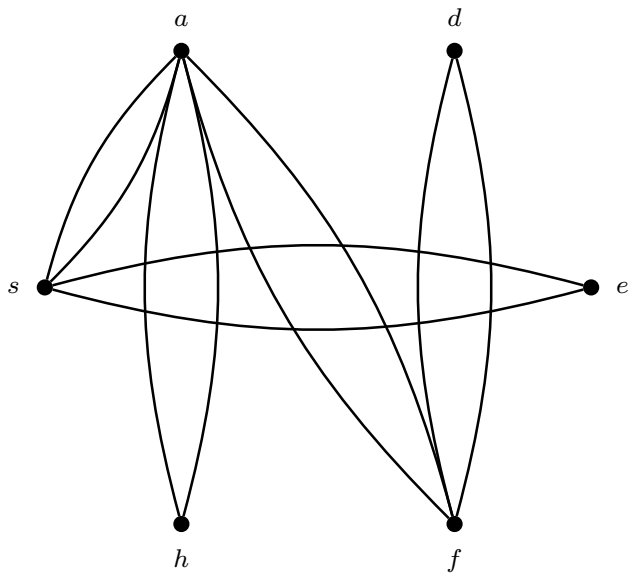
| | Austin | Dallas | El Paso | Fort Worth | Houston | San Antonio |
|---|---|---|---|---|---|---|
| Austin | . | 182 | 526 | 174 | 146 | 74 |
| Dallas | 182 | . | 568 | 31 | 225 | 253 |
| El Paso | 526 | 568 | . | 537 | 672 | 500 |
| Fort Worth | 174 | 31 | 537 | . | 237 | 241 |
| Houston | 146 | 225 | 672 | 237 | . | 189 |
| San Antonio | 74 | 253 | 500 | 241 | 189 | . |

*Solution.*

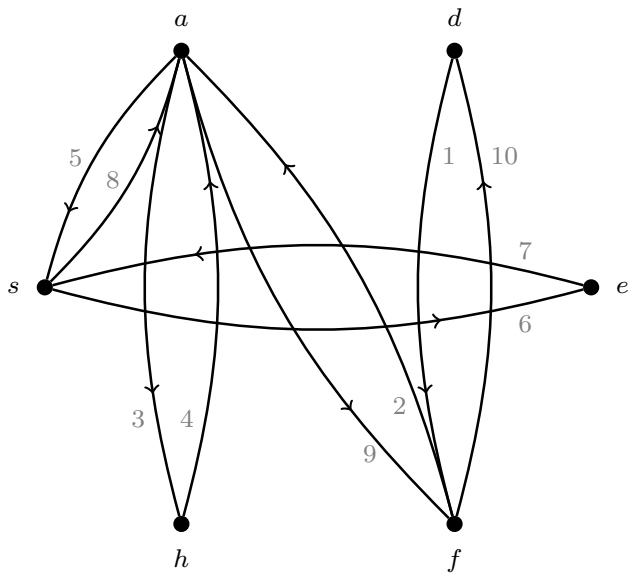
- Step 1. Find a minimum spanning tree (for example, by Prim's Algorithm).



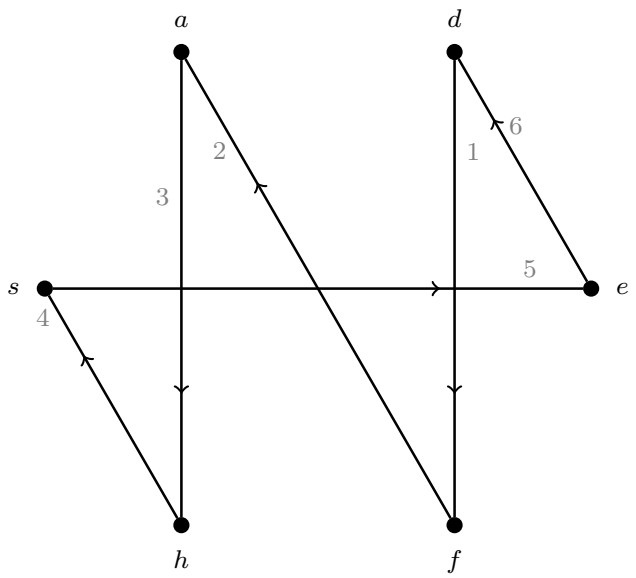

- Step 2. Duplicate all edges of the MST.




- Step 3. Find an Eulerian circuit.



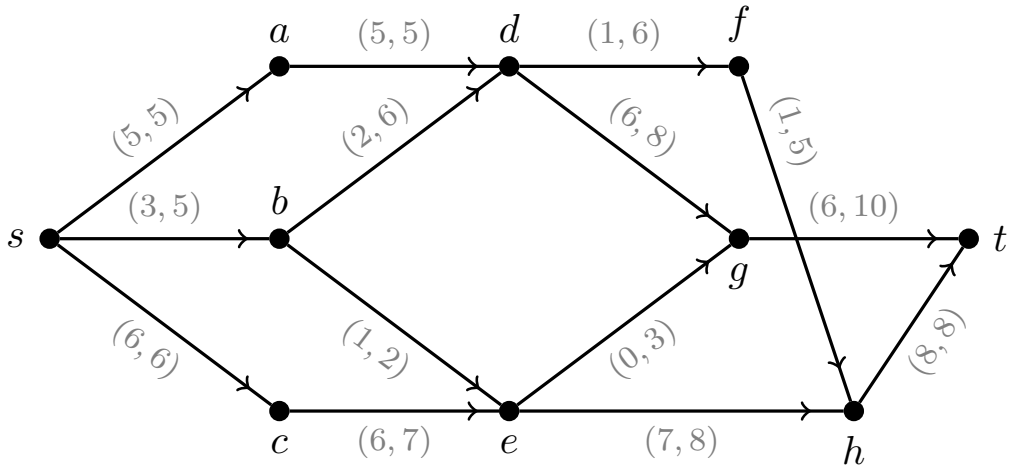

- Step 4. Convert the Eulerian circuit into a Hamiltonian cycle.



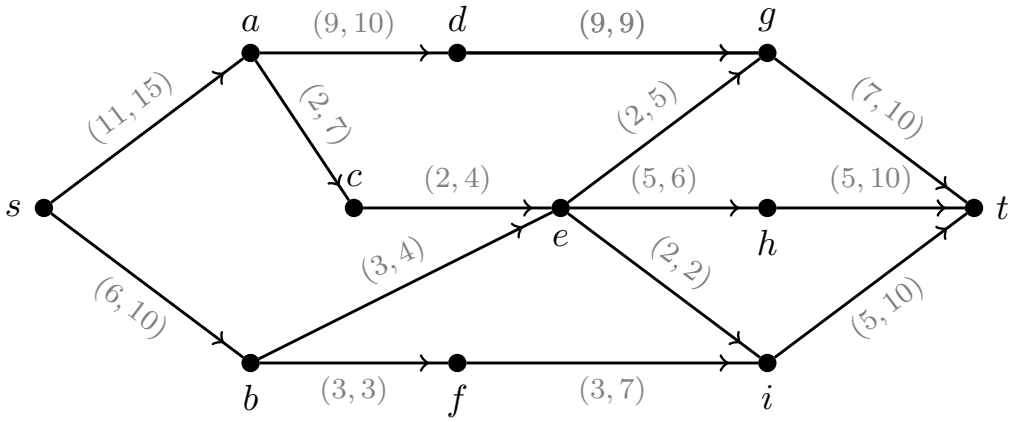
- Step 5. The total distance is 1608.

**Question 3.** For each of the networks below, use the Augmenting Flow Algorithm to maximize the flow and the Min-Cut Method to find a minimum cut.

1.




2.

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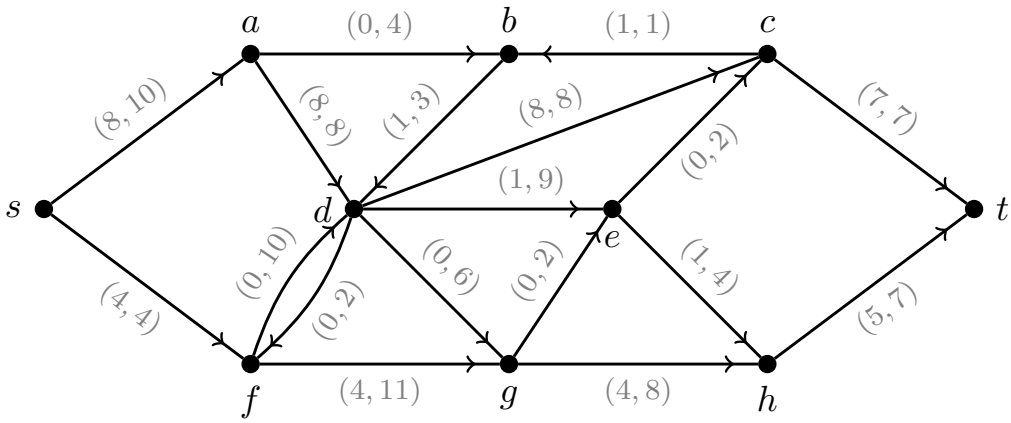
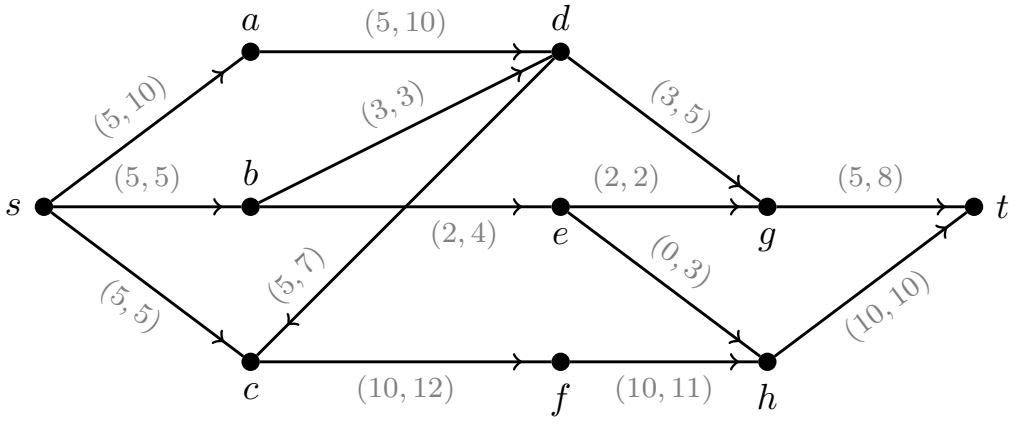
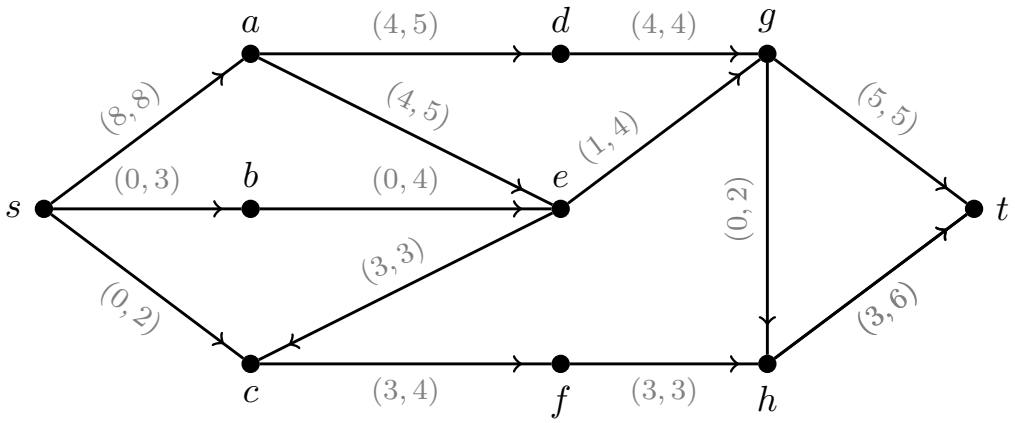


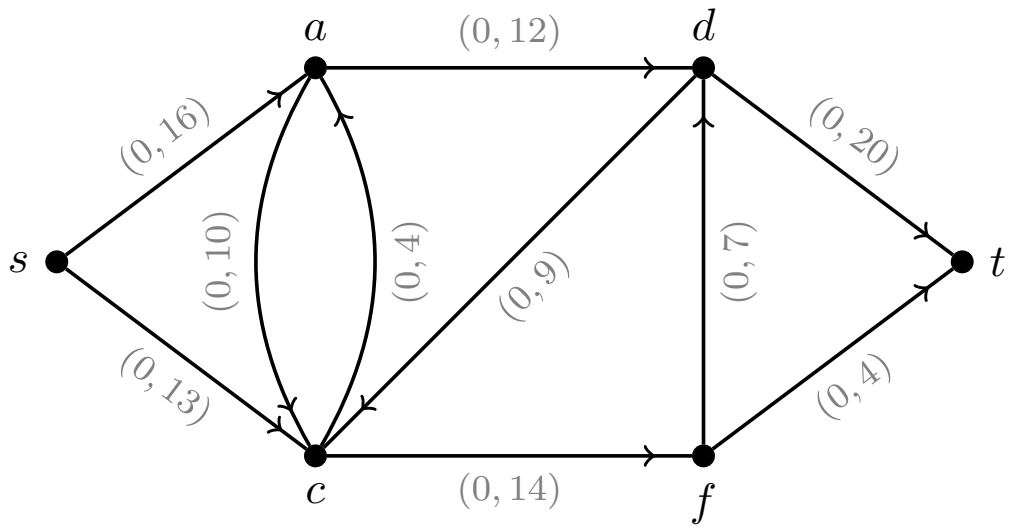

4.


a
(5, 10)
d
(5, 10)
(3, 3)
(3, 5)
(5, 5)
b
(2, 2)
(5, 8)
s
t
(2, 4)
e
g
(5, 7)
(0, 3)
(5, 5)
(10, 10)
c
(10, 12)
f
(10, 11)
h

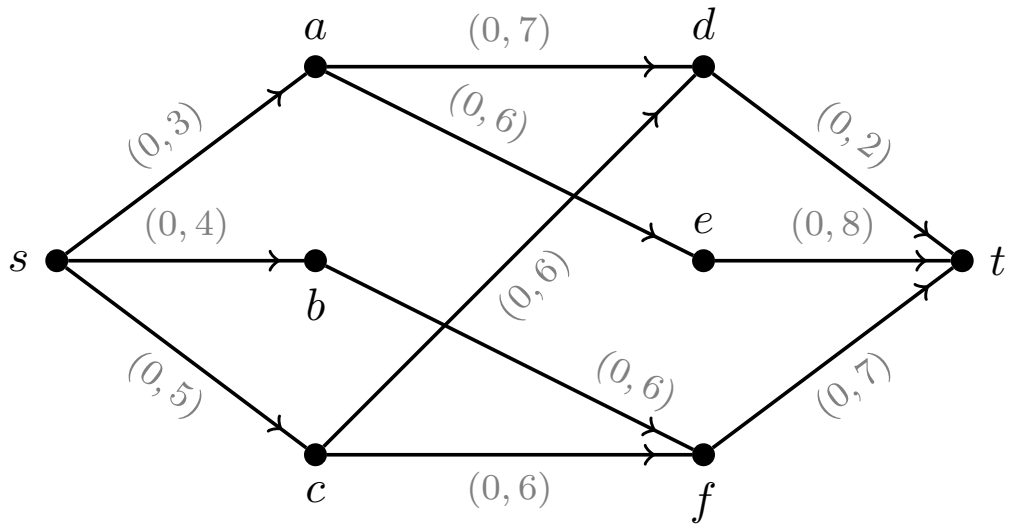
5.


a
(4, 5)
d
(4, 4)
g
(8, 8)
(4, 5)
(1, 4)
(5, 5)
(0, 3)
b
(0, 4)
e
(0, 2)
s
t
(3, 3)
(0, 2)
(3, 6)
c
(3, 4)
f
(3, 3)
h

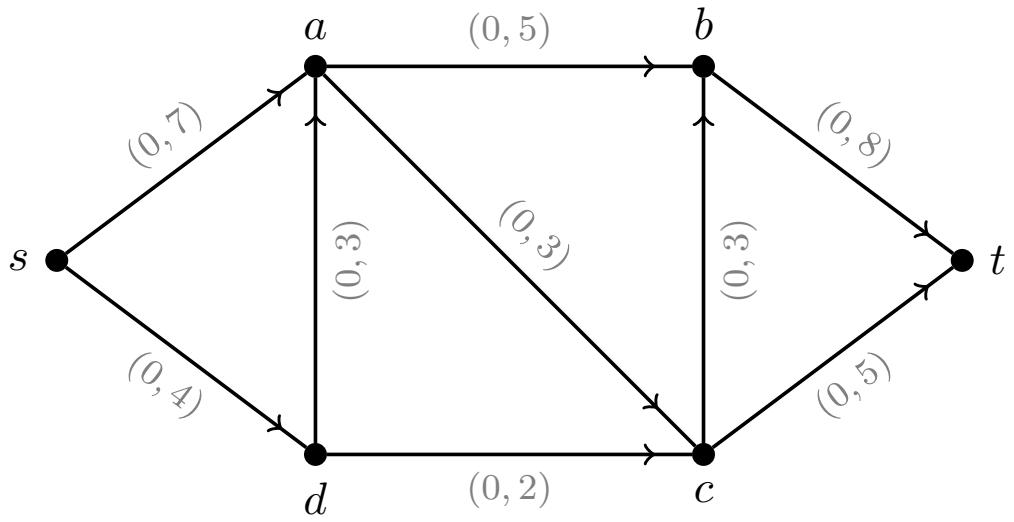
6.


a
(0, 12)
d
(0, 16)
(0, 20)
(0, 10)
(0, 4)
(0, 9)
(0, 7)
s
t
(0, 13)
(0, 4)
c
(0, 14)
f

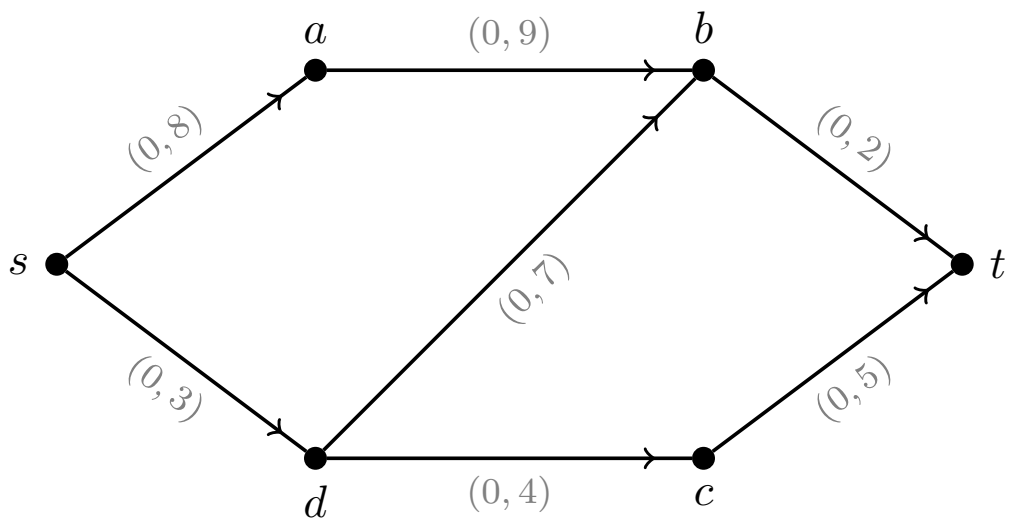
7.


a
(0, 7)
d
(0, 3)
(0, 6)
(0, 2)
(0, 4)
e
(0, 8)
s
t
b
(0, 6)
(0, 5)
(0, 6)
(0, 7)
c
(0, 6)
f

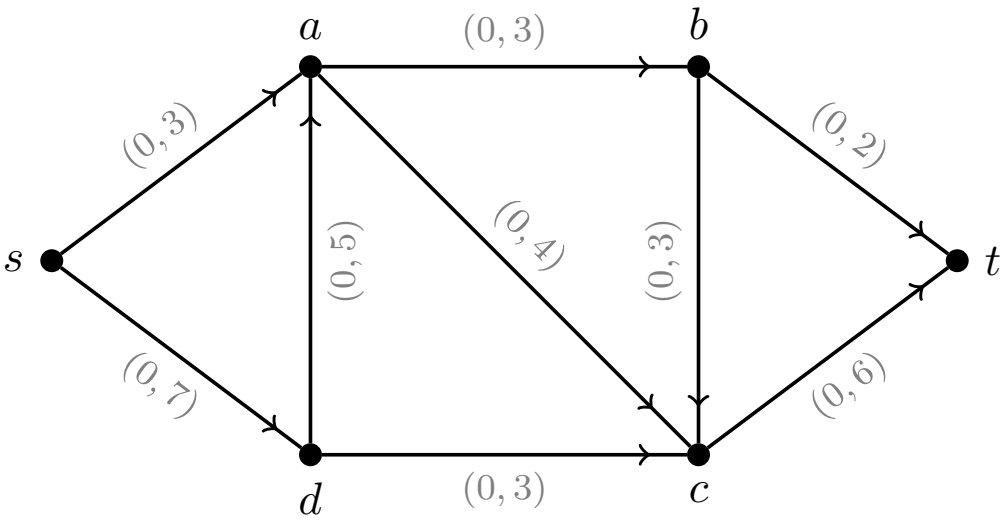
8.


a
(0, 5)
b
(0, 7)
(0, 8)
(0, 3)
(0, 3)
(0, 3)
s
t
(0, 4)
(0, 5)
d
(0, 2)
c

9.


a
(0, 9)
b
(0, 8)
(0, 2)
s
t
(0, 7)
(0, 3)
(0, 5)
d
(0, 4)
c

10.

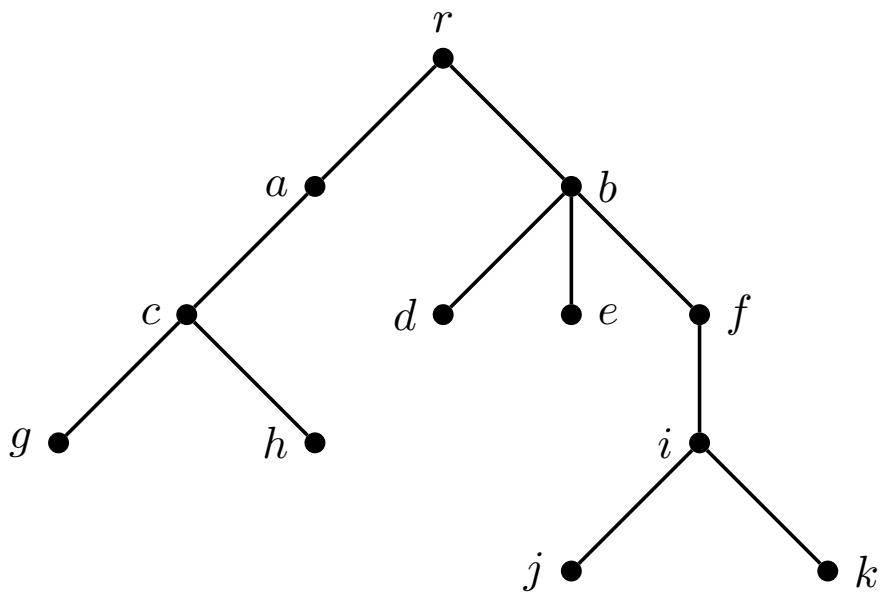



*Solution.*

1. Maximum flow $= 16$, $P = \{s\}$
2. Maximum flow $= 20$, $P = \{s, a, b, c, d\}$
3. Maximum flow $= 14$, $P = \{s\}$
4. Maximum flow $= 17$, $P = \{s, a, b, c, d, e, f\}$
5. Maximum flow $= 10$, $P = \{s, a, b, c, d, e, f, g\}$
6. Maximum flow $= 23$, $P = \{s, a, c, f\}$
7. Maximum flow $= 12$, $P = \{s, d, c, b, f\}$
8. Maximum flow $= 10$, $P = \{s, a, d\}$
9. Maximum flow $= 5$, $P = \{s, a, b\}$
10. Maximum flow $= 8$, $P = \{s, a, b, c, d\}$

**Question 4.** For the rooted tree $T$ below, with root $r$, identify the following:

1. The levels of $r$, $f$, $h$, and $k$.
2. The height of $T$.
3. The parents of $r$, $b$, $c$, $f$, and $i$.
4. The children of $r$, $b$, $c$, $f$, and $i$.
5. The ancestors of $r$, $a$, $d$, $h$, and $k$.
6. The descendants of $r$, $a$, $d$, $h$, and $k$.
7. The siblings of $a$, $f$, $h$, and $i$.




*Solution.*

1. The levels are $0, 2, 3$, and 4, respectively.
2. 4
3. The parents are: none; $r$; $a$; $b$; $f$, respectively.
4. The children are: $a, b$; $d, e, f$; $g, h$; $i$; $j, k$, respectively.
5. The ancestors are: none; $r$; $r, b$; $r, a, c$; $r, b, f, i$, respectively.
6. The descendants are: $a, b, c, d, e, f, g, h, i, j, k$; $c, g, h$; none; none; none, respectively.
7. The siblings are: $b$; $d, e$; $g$; none, respectively.

**Question 5.** Complete each of the following on the two graphs shown below.

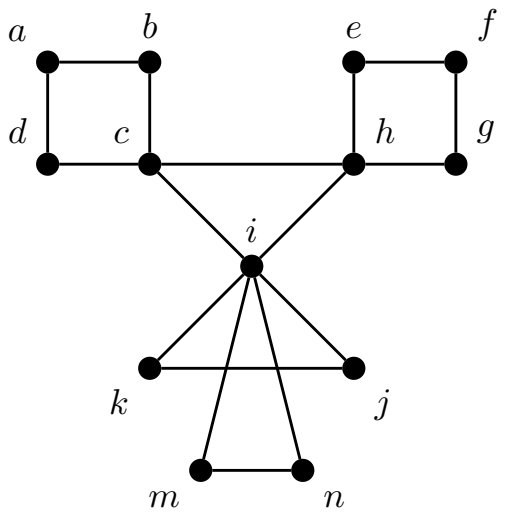
i) Find the breadth-first search tree with root $a$.

ii) Find the breadth-first search tree with root $i$.

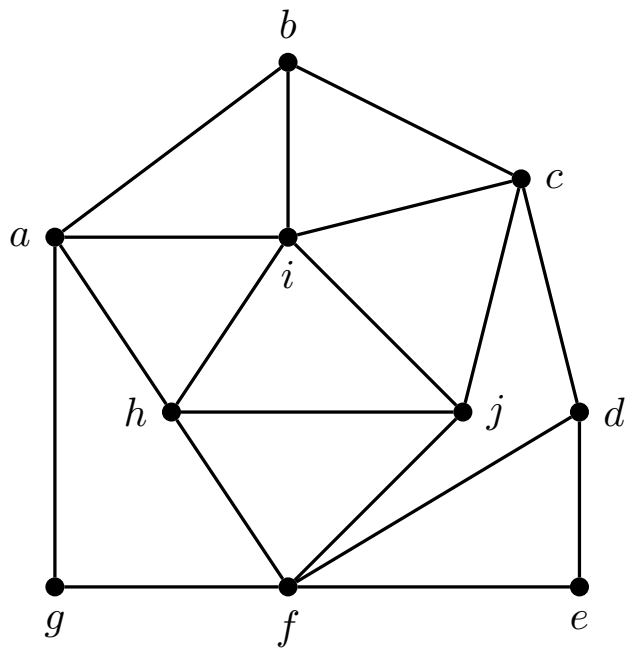
iii) Find the depth-first search tree with root $a$.

iv) Find the depth-first search tree with root $i$.

1.

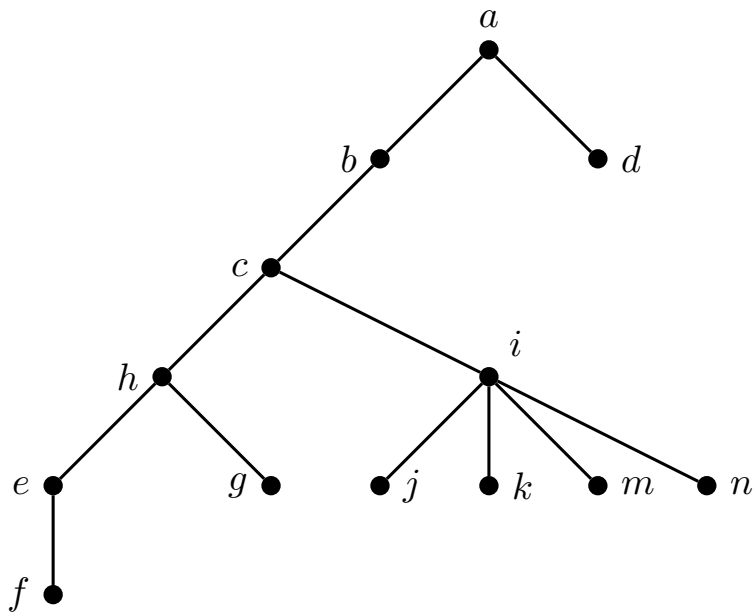



2.

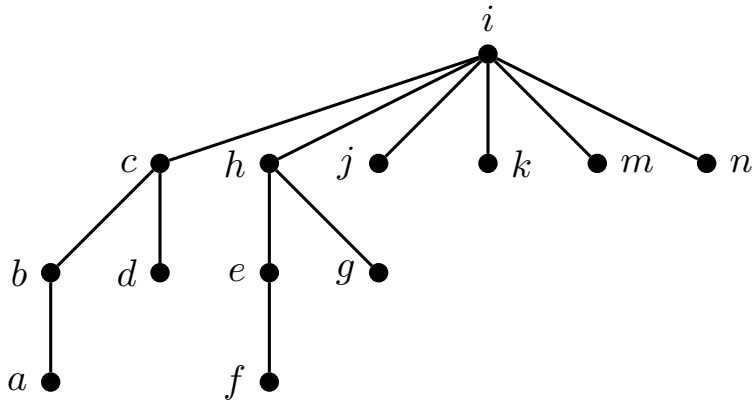



*Solution.*

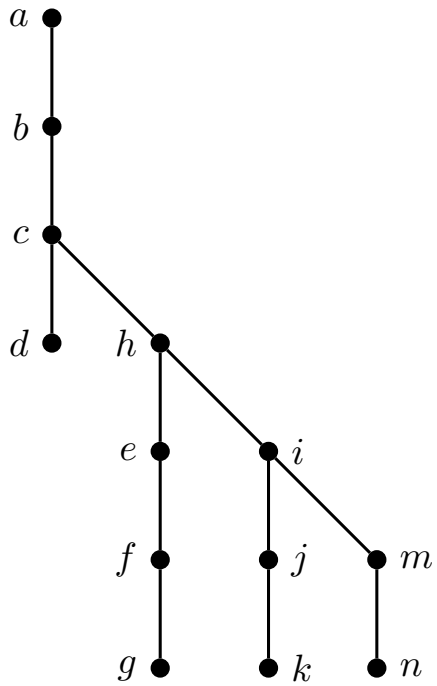
1. i) BFS tree with root $a$



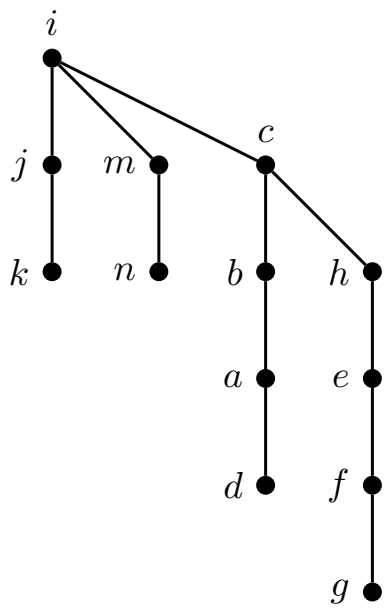

ii) BFS tree with root $i$



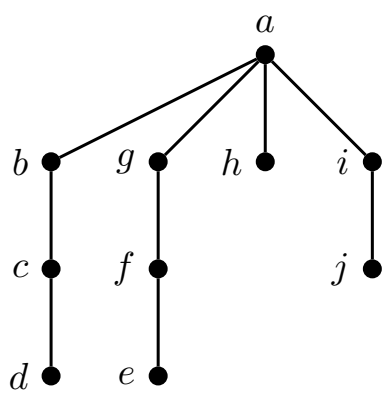

iii) DFS tree with root $a$



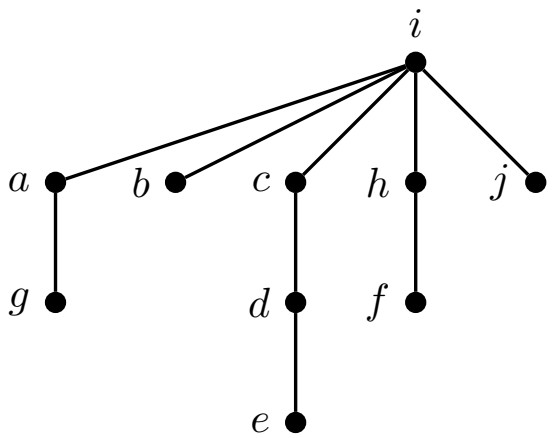
iv) DFS tree with root $i$



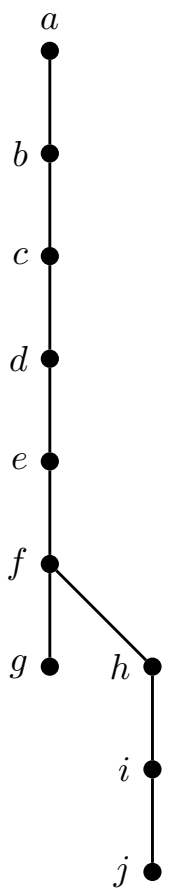

2. i) BFS tree with root $a$

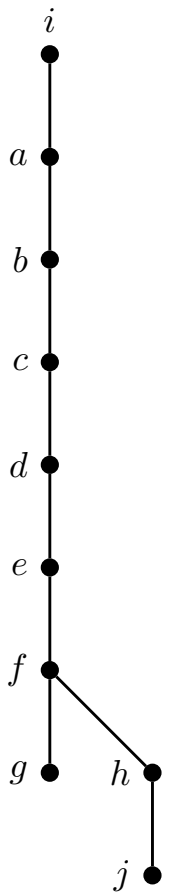



ii) BFS tree with root $i$




iii) DFS tree with root $a$




iv) DFS tree with root $i$